

ANSH BHARDWAJ

Data Engineer | Airflow Platform & Orchestration | GenAI & Agentic AI

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SUMMARY

Data engineer with 3 years owning the reliability of ~800 production Apache Airflow pipelines, the monitoring tooling around them, and the SQL and CI/CD automation that ships them. Pairs that platform depth with hands-on GenAI engineering — agentic AI, RAG, and multi-LLM orchestration — to automate data operations and software delivery.

EXPERIENCE

Accenture Solutions Pvt. Ltd. — Data Engineer · Sep 2023 – Present · Noida, India

Data Platform & Orchestration

- Own execution reliability for **~800 production Airflow DAGs**, covering scheduling, dependency design, failure triage, and SLA adherence.
- Built **Airflow monitoring and observability tooling** in Python over the **Airflow REST API** and **metadata database** to surface DAG health, failure patterns, and run-duration trends — then **re-architected it onto the REST API for Airflow 3** when direct metadata access was removed, preserving full coverage across a breaking upgrade.
- Authored reusable **AutoDAG templates** that standardized pipeline creation across the team and accelerated onboarding of new workflows.
- Developed **SQL-driven data workflows** and a **CI/CD automation framework**, including **Selenium regression suites** that validate UI integrity and **numerical accuracy** on a client-facing reporting portal carrying highly confidential figures — containerized in **Docker** and wired into **GitHub Actions** release gates.

GenAI & Agentic AI Engineering

- Built a **GenAI Airflow auto-healing framework** using LLMs to analyze pipeline failures, select remediations, and re-trigger DAGs, integrated with ServiceNow and Power Automate — **saving 250+ engineering hours per month and cutting team-wide MTTR by 20%**.
- Co-created **OmniSpark**, an enterprise **multi-LLM agentic AI platform** (Claude, OpenAI, Gemini) automating the full SDLC — Jira story authoring, feature branches, code generation, testing, code review, and PR creation — with an agent orchestration layer and **LLM policy gates** that keep autonomous changes reviewable and auditable.
- Co-developed **RAGENTS**, an internal VS Code extension offering a marketplace of reusable AI agents, engineering skills, prompts, and workflows.

PROJECTS

AI Research Agent — Local-First Agentic RAG System

Python · LangChain · LangGraph · Ollama · ChromaDB · FastAPI

- Built a fully local **RAG system** — Ollama inference, sentence-transformers embeddings, ChromaDB vector store — with multi-format ingestion and **incremental indexing** at zero API cost.
- Orchestrated a **4-node LangGraph state machine** (researcher → analyst → critic → writer) over typed state, plus a **tool-calling agent** that chooses per query between the private index and live web.
- Served it over **FastAPI with token streaming (SSE/NDJSON)**, cutting time-to-first-token via cached model clients, per-node output budgets, and context clipping.

TECHNICAL SKILLS

Programming: Python, SQL

Data Engineering: Apache Airflow (2 & 3), Airflow REST API & Metadata DB, Workflow Orchestration, DAG & Dependency Design, Pipeline Monitoring & Observability, Incremental Processing, Data Governance

GenAI: Agentic AI, RAG, Multi-Agent Orchestration, Tool-Calling Agents, Vector Embeddings, Semantic Search, Prompt Engineering, Multi-LLM Architecture, LLM Policy Gates

Tooling: Claude / OpenAI / Gemini APIs, LangChain, LangGraph, Ollama, ChromaDB, FastAPI, Copilot Studio, Power Automate, Docker, Git, GitHub Actions, CI/CD, REST APIs, Jira, ServiceNow

CERTIFICATIONS

Microsoft **DP-203** Azure Data Engineering · Anthropic **Claude Certified Architect — Professional** · Anthropic Claude Certified (CCA-F) · Astronomer Airflow Fundamentals · Astronomer DAG Monitoring & Auditing · Microsoft GitHub GHE-300

EDUCATION & AWARDS

B.Tech — ABES Engineering College · 2023

ACE Award, Accenture (2026) · MC Star Award & SOS Award — Client Recognition